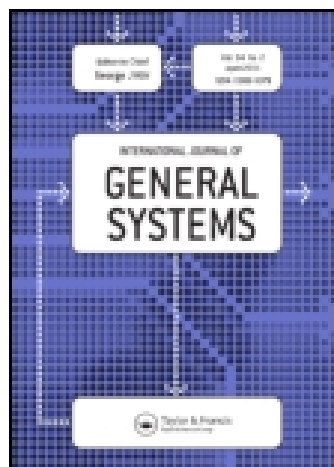




International Journal of General Systems

Publication details, including instructions for authors and subscription information:

<http://www.tandfonline.com/loi/ggen20>

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Published online: 17 Nov 2011.

To cite this article: Plamen Angelov & Ronald Yager (2012) A new type of simplified fuzzy rule-based system, International Journal of General Systems, 41:2, 163-185, DOI:

[10.1080/03081079.2011.634807](https://doi.org/10.1080/03081079.2011.634807)

To link to this article: <http://dx.doi.org/10.1080/03081079.2011.634807>

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A new type of simplified fuzzy rule-based system

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(Received 23 October 2010; final version received 20 October 2011)

In memoriam: We dedicate this paper to Professor Abe Mamdani (Imperial College, London, England) who recently passed away. He left one of the brightest results in this area and we are indebted to the Pioneers like him

Over the last quarter of a century, two types of fuzzy rule-based (FRB) systems dominated, namely Mamdani and Takagi–Sugeno type. They use the same type of scalar fuzzy sets defined per input variable in their antecedent part which are aggregated at the inference stage by t-norms or co-norms representing logical AND/OR operations. In this paper, we propose a significantly simplified alternative to define the antecedent part of FRB systems by data *Clouds* and density distribution. This new type of FRB systems goes further in the conceptual and computational simplification while preserving the best features (flexibility, modularity, and human intelligibility) of its predecessors. The proposed concept offers alternative non-parametric form of the rules antecedents, which fully reflects the real data distribution and does not require any explicit aggregation operations and scalar membership functions to be imposed. Instead, it derives the fuzzy membership of a particular data sample to a Cloud by the data density distribution of the data associated with that Cloud. Contrast this to the clustering which is parametric data space decomposition/partitioning where the fuzzy membership to a cluster is measured by the distance to the cluster centre/prototype ignoring all the data that form that cluster or approximating their distribution. The proposed new approach takes into account fully and exactly the spatial distribution and similarity of all the real data by proposing an innovative and much simplified form of the antecedent part. In this paper, we provide several numerical examples aiming to illustrate the concept.

Keywords: fuzzy rule-based systems; Mamdani and Takagi–Sugeno fuzzy systems; recursive least square estimation; data density and distribution; clustering

1. Introduction

During the last four decades, the fuzzy sets and fuzzy rule-based (FRB) systems emerged and are widely accepted as a dominant mechanism and framework to capture and to represent intelligent systems (systems that have elements of reasoning and certain level of intelligence). Two of the three main types of FRB systems [the so-called Mamdani (Zadeh 1973, Mamdani and Assilian 1975) or Zadeh–Mamdani and Takagi–Sugeno (TS 1985) type] gained more prominent attention and wider application. The other main type of FRB systems (relational; Pedrycz 1983) is less popular due to conceptual and computational difficulties. Comparing these two types, there are notable similarities (they both share

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exactly the same type of antecedent/premise part which is scalar fuzzy-sets-based). They differ by their consequents part which for the TS type is of crisp, functional type while for the Mamdani type is of fuzzy-sets-based.

The antecedent part is determined by a number of fuzzy sets (one per each variable), which are themselves defined by parameterized scalar membership functions. These membership functions are determined either by experts (an approach used predominantly in the 1970–1980s and less so now) or from data (a popular approach from 1990s). There are number of issues with such an approach, including,

- (i) The degree of activation of a fuzzy rule is determined as an aggregation of the degrees of membership of a data sample to each of the fuzzy sets [at least two different approaches are widely used for aggregation called t-norms – minimum and product – but there are a number of other, less popular (Klir and Folger 1988) ones].
- (ii) Defining a membership function requires parameterization – determining the centre and left/right boundaries or spread (if Gaussian or bell-shaped function is used);
- (iii) Membership functions often differ significantly from the real data distribution.

In this paper, we propose an entirely new concept to the way the antecedent part is defined. Based on this, a new simplified type of FRB is proposed as an alternative to both Mamdani and TS types of FRB. According to the proposed concept, the system is assumed to be decomposable into a set of loosely connected local simpler (linear, singleton, exponential, etc.) systems aggregated in a fuzzy way. Each local sub-system, however, is valid for a certain sub-set of the entire data set only, which is called a data Cloud. This concept can be seen as an extension of the well-known concepts of the case-base reasoning (Watson 1999) and k-nearest neighbours (Hastie *et al.* 2001) but with a much more sophisticated mathematical underpinning being computationally and conceptually richer (it assumes fuzzy membership of a data sample to more than one Cloud at the same time with different degree of association/membership determined by the local density to *all* samples from that Cloud). It can also be seen as ‘going back to the roots’ of fuzzy sets concept as defined by Zadeh (1973) in the sense that it concentrates on the comparing objects rather than comparing features of objects (scalar variables). It removes the problems related to the membership functions definition and representation in a parametric form. In this sense, it resembles popular recently non-parametric particle filters (Arulampalam *et al.* 2002) where non-Gaussian distributions are considered, but the technique proposed in this paper is applicable on-line and in real-time since it is recursive and one pass.

The proposed approach replaces the scalar (per variable) membership functions with a non-parametric function, which represents the local (per Cloud) data density. In this respect, it has some resemblance with the other well-known kernel-based approaches such as Parzen windows (Hastie *et al.* 2001) and support vector machines (Vapnik 1998). The intention is to simplify the FRB definition by removing the problems related to the definition of scalar parameterized membership functions. In the new concept, there is no need to define centres/prototypes/focal points of the fuzzy sets.

The similarity/dissimilarity is closely linked with the notion of distance. In the proposed approach, there is no specific requirement to use Euclidean type of distance (alternatives such as Mahalanobis, cosine, or any other are also acceptable). The proposed concept touches the very foundations of the complex systems identification and thus its application domain ranges from simple clustering-based techniques for pattern recognition, image segmentation, vector quantization, etc., to more general modelling, prognostics, classification, and time-series prediction problems in various application

Table 1. Types of FRB and their differences.

	<i>Antecedent/ IF part</i>	<i>Consequent (THEN) part</i>	<i>De-fuzzification</i>
Mamdani	Scalar, parameterized	Scalar, parameterized	Centre of gravity
TS	fuzzy sets	fuzzy sets	
Proposed new type	All data non-parametric	Functional (usually	Fuzzily weighted
FRB system	data Clouds	linear)	sum (average)

areas, e.g. intelligent sensors, mobile robotics, advanced manufacturing processes, sensor networks, etc. Several numerical examples are presented primarily as a proof of concept and more applications will be presented in future publications.

2. The concept and structural framework of the proposed method

Comparing the two traditional types of FRB systems (see Table 1), one can observe their similarity in terms of the antecedent (premise) part.

While both the consequent part and the defuzzification inference differ, the antecedent part of both is exactly the same. Yet, this type of antecedent part formulation is often a stumbling block in practical design of FRB systems. This is true both in the case when their design relies on real data as well as when it relies on expert knowledge. The reason is that defining membership functions per scalar variable and parameterization of all of them requires a very high level of approximation (because the real data distributions and real problems are often not smooth and easy to describe ‘per variable’). Addressing this important bottleneck of the FRB systems design and interpretation, we propose a simplified and effective new form of antecedent/premise part which makes the overall FRB intrinsically generic multi-input multi-output (MIMO) modelling framework that covers various types of systems including but not limited to fuzzy rules and neural networks (NNs), see Figure 1. Note that the NN interpretation of the proposed approach is simpler than the respective TS type neuro-fuzzy systems such as ANFIS (Jang 1993), DENFIS (Kasabov and Song 2002), eTS + (Angelov and Zhou 2008a; Angelov 2010), SAFIS (Leng *et al.* 2002), FLEXFIS (Lughofer 2008), ePL (Lima *et al.* 2006), and SOFNN (Rong *et al.* 2006) having fewer layers and parameters.

Let us consider a complex, generally non-linear, non-stationary, non-deterministic system that can only be described and observed by its input and output vectors $x = [x_1, x_2, \dots, x_n]^T$ and $y^i = [y_1^i, y_2^i, \dots, y_m^i]$, respectively. The aim is to describe the input–output dependence based on a history of observation of input–output pairs, $z_j = [x_j^T; y_j^T]^T$, $j = 1, 2, \dots, k-1$ and current, k inputs, x_k^T only. The dimension of the vector of input–output data z_j is $(n+m)$: n dimensions of the inputs and m dimensions of the outputs.

Traditional FRB systems that address such problem include

$$\text{Mamdani : Rule}^i : \text{IF (ant}^i) \text{ THEN (} y^i \text{ is } \text{LT}_{n+1}^i \text{)}, \quad (1a)$$

$$\text{TS : Rule}^i : \text{IF (ant}^i) \text{ THEN (} y^i = x_e^T \pi^i \text{)}, \quad (1b)$$

where Rule^i denotes the i th fuzzy rule; LT_j^i , $i = [1, N]$, $j = [1, n]$ denotes the j th linguistic term (e.g. small, medium, large, etc.) for the i th fuzzy rule; N is the overall number of fuzzy rules; y denotes the output variable; π denotes the vector of parameters, $\pi^i = [a_0^i \ a_1^i \ a_n^i]^T$; and $x_e^T = [1, x^T]$ denotes the extended inputs vector.

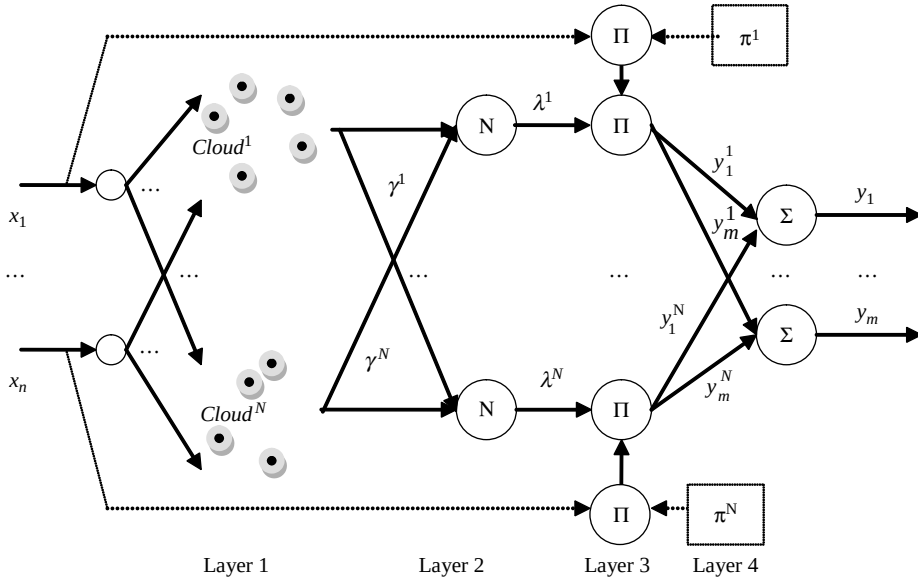


Figure 1. A MIMO structure of the proposed simplified FRB in a NN form.

In both cases, the antecedent of the i th rule is described as

$$\text{ant}^i : ((x_1 \text{ is } \text{LT}_1^i) \dots \text{AND}(x_n \text{ is } \text{LT}_n^i)), \quad (1c)$$

where $x_j; j = [1, n]$ denotes the j th input variable.

The aggregation of the contributions of all fuzzy rules to the overall output is usually done in Mamdani type FRB by a so-called centre-of-gravity defuzzification (Klir and Folger 1988, Yager and Filev 1994) and in TS type FRB by fuzzily weighted averaging.

The new alternative proposed in this paper is based on data Clouds formed using recursively calculated relative density in the input–output data space. These Clouds are then used as building blocks of the fuzzy rules. Despite some resemblance to the clustering, there are several major differences between Clouds and clusters, see Figure 2.

The main difference is that the proposed approach does not consider and does not require membership functions or fuzzy sets per scalar variable to be formulated. In this sense, the proposed simplified FRB structure can also be seen as type 0 fuzzy sets (by analogy to the type II fuzzy sets for which the membership functions are defined by a fuzzy set for each point of the membership function; in contrast, the proposed approach *does not require* an explicit definition of the membership function or even a prior assumption of its form).

A very interesting and strong aspect of the proposed method is the *non-parametric* form of the data Clouds as local building blocks of the overall complex system.

Data Clouds are sets of previous data samples with common properties (closeness in terms of the input–output mapping). They *directly* represent *all previous data samples*. In contrast to this, the traditional membership functions usually do *not* represent the true data distributions; instead, they represent some desirable/expected/estimated (often subjectively) preferences. The fuzziness of the proposed method is preserved in the manner of decomposition in the sense that a particular data sample can belong to all Clouds with different degree, $\gamma \in [0, 1]$. Importantly, Clouds *do not have or require boundaries* and,

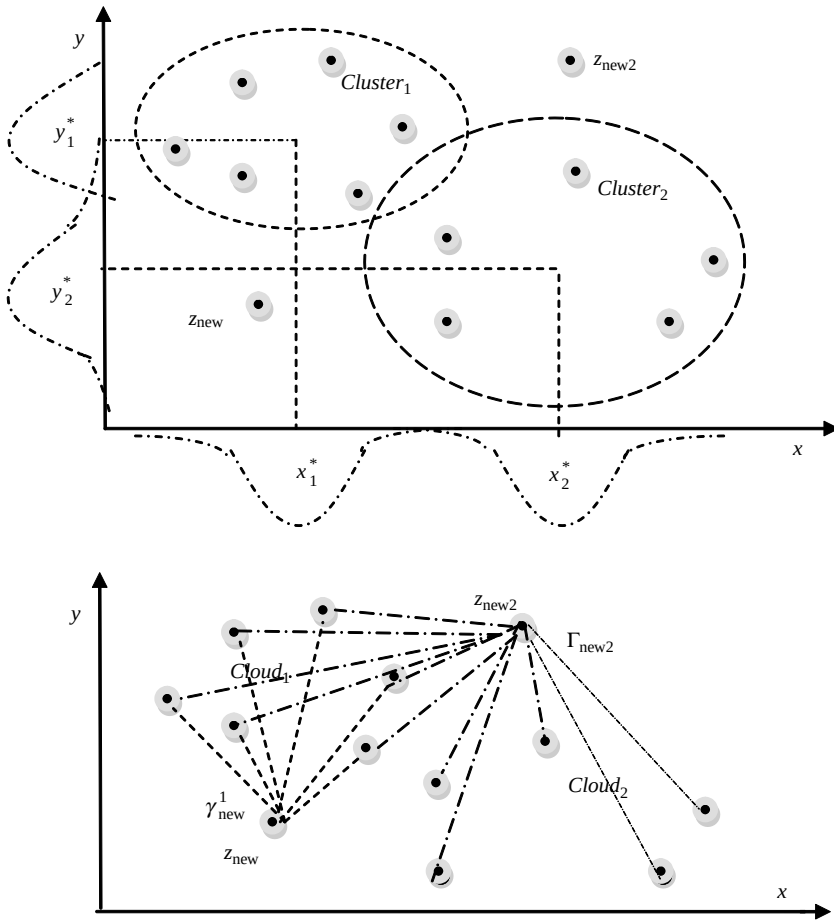


Figure 2. Top, the traditional partitioning through clustering and parameterized scalar membership functions; bottom, the proposed approach [local (γ)] and global densities (Γ) are illustrated. Note that there are no boundaries or specific shapes associated with the Clouds.

thus, they do not have specific shapes. A Cloud is described by the set of data samples that belong to it and linguistically by a statement of the following form:

$$(z \text{ is like } \mathfrak{Z}^i), \quad (2a)$$

where $\mathfrak{Z}^i, \mathfrak{Z} \in R^{n+m}, i = [1, N]$ denotes a Cloud in the input-output data space (subset of real input-output data with similar properties).

$$(x \text{ is like } \mathfrak{N}^i), \quad \mathfrak{N}^i \in R^n, \quad i = [1, N], \quad (2b)$$

where $\mathfrak{N}^i, \mathfrak{N} \in R^n, i = [1, N]$ denotes a Cloud in the inputs only data space (subset of real input data with similar properties).

The degree of membership to a Cloud is measured by the normalized [using fuzzily weighted average (Klir and Folger 1988, Yager and Filev 1994)] local density for a

particular data sample, x_k :

$$\lambda_k^i = \frac{\gamma_k^i}{\sum_{j=1}^N \gamma_k^j}, \quad i = [1, N], \quad (3)$$

where γ^i is the *local* density of the i th Cloud for a particular data sample, which is defined by a suitable kernel over the distance between the current sample, x_k , and *all* the other samples *from that* Cloud (therefore *local*),

$$\gamma_k^i = K \left(\sum_{j=1}^{M^i} d_{kj}^i \right), \quad i = [1, N], \quad (4a)$$

where M^i denotes the number of input data samples associated with the i th Cloud.

Similarly, *global* density Γ for a particular data sample, z_k , which is defined by a suitable kernel over the distance between the current input–output sample, z_k , and *all* the other input–output samples (therefore *global*),

$$\Gamma_k = K \left(\sum_{j=1}^k d_{kj} \right), \quad (4b)$$

Different types of distance measures can be used [(each having its own advantages and disadvantages (Angelov and Zhou 2008a)]. For example, one can use Euclidean distance, $[d_{kj}^i]_E^2 = \|x_k - x_j\|^2$ or $[d_{kj}]_E^2 = \|z_k - z_j\|^2$, cosine distance, $[d_{kj}]_{\cos} = z_k z_j / \|z_k\| \|z_j\|$, etc.

For problems such as classification, the weighted average (Equation (3)) may be replaced with the so-called ‘winner takes all’ inference operator (Klir and Folger 1988, Yager and Filev 1994) giving more prominence to the most relevant Cloud. For prediction, systems modelling and control, weighted average is preferred inference (Yager and Filev 1994). The kernel (Aizerman *et al.* 1964) is a well-known measure of similarity and Cauchy type of function is specifically interesting (Angelov and Buswell 2002). The local density with a Cauchy type of function can be defined as

$$\gamma_k^i = \frac{1}{1 + (1/M^i) \sum_{j=1}^{M^i} (d_{kj}^i)^2} = \frac{1}{1 + (\bar{d}^2)_k^i}, \quad (5)$$

where $\bar{d}$ denotes the mean/average distance from the current, k th point to *all* the points of the i th Cloud.

It can be proven that the Cauchy type function asymptotically tends to Gaussian, but can be calculated recursively (Angelov 2011):

$$\gamma_k^i = \frac{1}{1 + \|x_k - \mu_k^L\|^2 + \sum_k^L - \|\mu_k^L\|^2}, \quad (6)$$

where $\mu_k^L = ((M_k^i - 1)/(M_k^i))\mu_{k-1}^L + 1/(M_k^i)x_k$, $\mu_1^L = x_1$, is the *local* mean value of the data of that Cloud,

$$\sum_k^L = \frac{M_k^i - 1}{M_k^i} \sum_{k-1}^L + \frac{1}{M_k^i} \|x_k\|^2, \quad \sum_1^L = \|x_1\|^2.$$

In a much similar way, the global density, Γ_k , can be defined where the only difference is the way the mean and variance are calculated – now they concern *all* the points instead of points form a specific Cloud:

$$\Gamma_k = \frac{1}{1 + (1/(k-1)) \sum_{j=1}^{k-1} d_{kj}^2}, \quad (7)$$

where $\mu_k = ((k-1)/k)(\mu_{k-1}) + (1/k)x_k$, $\mu_1 = x_1$, is the *global* mean value of *all* data available at that moment, k .

$$\Sigma_k = \frac{k-1}{k} \Sigma_{k-1} + \frac{1}{k} \|x_k\|^2, \quad \Sigma_1 = \|x_1\|^2.$$

It is easy to check that because of the way Equation (3) is formulated, the degree of fuzzy membership to a Cloud, λ^i , is normalized, that is,

$$\sum_{i=1}^N \lambda^i = 1. \quad (8)$$

We can now define the simplified FRB as

$$\text{Rule}^i : \text{IF}(x \text{ is like } \aleph^1) \text{ THEN}(y^i = x_e^T \pi^i), \quad (9)$$

where the degree of fulfillment of the premise part is determined by the local density, γ^i , and

$$\pi^i = \begin{bmatrix} a_{01}^i & a_{02}^i & \cdots & a_{0m}^i \\ a_{11}^i & a_{12}^i & \cdots & a_{1m}^i \\ \cdots & \cdots & \cdots & \cdots \\ a_{n1}^i & a_{n2}^i & \cdots & a_{nm}^i \end{bmatrix}$$

are the consequent sub-system parameters (in this MIMO system, the output is m -dimensional, i.e. $y \in R^m$).

The overall output of the proposed simplified FRB system, y (see Figure 1), is formed as a collection of loosely/fuzzily combined multiple locally (per Cloud) valid simpler sub-models, y_i :

$$y = \sum_{i=1}^N \lambda^i y^i, \quad (10)$$

where y^i represents the output of the i th local sub-system.

The simplified FRB as described by Equations (3), (4), (9) and (10) can be graphically represented as a four-layer feed-forward NN as illustrated in Figure 1. The first layer is quite different from the neuro-fuzzy systems like ANFIS (Jang 1993), DENFIS (Kasabov and Song 2002), eTS+ (Angelov 2010), SAFIS (Leng *et al.* 2002), FLEXFIS (Lughofer 2008), ePL (Lima *et al.* 2006), and SOFNN (Leng *et al.* 2002). No scalar parameterized membership functions are defined in the proposed approach; instead, a given input data sample, x_k , is compared in a recursive computationally efficient way to *all* previous data samples *per Cloud* and the *local* density of each Cloud in terms of this data sample is calculated by (6). The second layer of the network takes as inputs the density of the

respective Cloud, γ^i , and gives as output the normalized firing level of the fuzzy rule (which is the membership to the i th Cloud), λ^i using (3). The first two layers represent the antecedent part of the fuzzy rules; note that this representation is simpler than for Mamdani and TS types of FRB systems [in ANFIS, DENFIS, eTS+, SAFIS, FLEXFIS, ePL, SOFNN, etc., for example, there are three layers which produce the normalized firing strength (activation level) of a particular rule, λ^i]. The third layer aggregates the antecedent and the consequent part that represents the local sub-systems (singletons or hyper planes). Finally, the last layer forms the total output of the simplified FRB system. It performs a weighed summation of local sub-systems according to Equation (3).

3. Complex systems identification through density-based Clouds

3.1 Structure identification

Having described the structure of the proposed simplified FRB system, its design will now be described. Essentially, the identification of any system consists of two basic parts (Ljung 1999):

- (i) structure identification and
- (ii) parameter identification.

Traditionally, FRB systems were initially (until mid-1980s) designed using the so-called ‘domain expert knowledge’ represented as a set of linguistic fuzzy rules (Zadeh 1973, Mamdani and Assilian 1975, Klir and Folger 1988). This approach has still some attractiveness in decision support systems. During the 1990s, the fact that there is a huge amount of data available in various types of applications lead to the development of the so-called ‘data-driven’ or ‘data-centred’ techniques (Takagi and Sugeno 1985, Jang 1993, Yager and Filev 1994, Babuska 1998) which borrowed heavily from machine learning. The last decade is marked by intensive development of yet more sophisticated techniques, which are called ‘knowledge extraction from data streams’ (Angelov and Buswell 2002, Angelov and Zhou 2008a, Angelov 2011), introducing the so-called concept of ‘system structure evolution’ and ‘evolving fuzzy systems’ (Angelov and Zhou 2008b, Angelov 2010).

In the literature, the problem of system structure identification was traditionally left to the choice of the system designer. This problem was paid much more attention since Mountain clustering (Yager and Filev 1993) was proposed to be used automatically to solve the problem of FRB systems design. Later, its modified version known as subtractive clustering (Chiu 1994) and, more notably, the concept of *system structure evolution* (Angelov and Buswell 2002, Angelov 2010, 2011) further developed this design technique. The problem of parameter optimization has been traditionally more widely developed (Jang 1993, Yager and Filev 1994, Ljung 1999, Kailath *et al.* 2000). We propose to ‘discover’ the underlying structure of a complex system based on data Clouds determined using the data density. As we stressed in the previous section and in Figure 2, the *Clouds* differ from clusters in several aspects, which are summarized in Table 2.

The proposed approach is governed by the following main principles:

- (A): good *generalization and summarization* of the input–output relation/mapping – this is achieved by forming new Clouds from data samples which have high global density, Γ ;
- (B): avoid *excessive overlap*, *old* Clouds, or the ones that are *rarely utilized* and
- (C): Maintain the quality of the Clouds on-line and remove irrelevant or outdated Clouds

Table 2. Clouds vs clusters.

Aspect	Clustering	Granulation
Boundaries	Defined	No boundaries
Centre/prototype	Defined	None
Distance to	Centre/prototype	All data (accumulated)
Membership functions	Scalar	Vector
	Parameterized	Non-parametric

In this paper, without limiting the applicability of the overall concept (off-line, on-line, and evolving) to a particular type of forming the Clouds, we perform granulation in a dynamically evolving manner quite similar to the recently introduced eClustering approach (Angelov 2010). In addition, we propose and demonstrate a simple yet effective approach for classification using one rule per class data Clouds and density distribution-based simplified FRB.

3.1.1 Evolving simplified FRB predictor/estimator

The dynamically evolving FRB addressing the prediction and estimation problems forms new Clouds (evolves the structure of the FRB system) starting either from an initially existing structure (this may be designed off-line or suggested by an expert) or, if such initial structure does not exist, ‘from scratch’. Let us assume ‘starting from scratch’, because this is the more general and more challenging case. The very first data sample (in the case of classification problem, the very first sample per class), naturally, starts the formation of the first Cloud ($i = 1$). For all next input–output data samples [note that in prediction when predicting k th output we will use the structure determined based on $k - 1$ input–output data samples in a manner typical for estimation and control theories (Ljung 1999, Kailath *et al.* 2000)], there are essentially two possible scenarios:

- (1) they are associated with the existing Clouds updating the local density of the nearest one, and
- (2) they initiate a new Cloud if principle (A) above requires this.

The first one is obvious and it invokes the update of Equation (6). The second case concerns input–output data samples for which the global density calculated at these points is higher than the global density estimated at the initial points of all existing Clouds:

$$\Gamma_k > \Gamma_k^i, \quad \forall i | i = [1, N]. \quad (11)$$

Note that a new Cloud is initiated ($z_k \rightarrow z^*$) when condition (11) is satisfied for *all* existing Clouds ($\forall i$). Such cases are not very often.

Finally, we check if principle (B) is satisfied by checking for each data sample which is a candidate to start a new Cloud (one that satisfies (11)) if this data samples satisfy the so-called ‘one sigma’ condition (Hastie *et al.* 2001):

$$\exists i; i = [1, N], \quad |\gamma_k^i| > e^{-1}. \quad (12)$$

If this condition is satisfied, a new Cloud is NOT formed even if condition (11) is satisfied. The other aspects of condition (B) such as *age* and *utility* of the Cloud will be described in

the next section and are similar to the ones used in advanced clustering (Angelov and Zhou 2008a, Angelov 2010).

3.1.2 One rule per class simplified FRB classifier

We will demonstrate the simple FRB method with a classifier that has a single rule per class. That means we assume that all the data of a given class form a single data Cloud (in a more general case one can have more than one Cloud per class either in an off-line manner or evolving them from data). The aim is to design a simple FRB classifier of the following type:

$$\text{Rule}^i : \text{IF}(x \text{ is like Cloud}^i) \text{ THEN}(x \rightarrow \text{Class}^i), \quad i = [1, C], \quad (13)$$

where C is the number of classes and Class^i denotes the label of the i th class.

This FRB classifier will always have exactly C fuzzy rules and the antecedent of each rule will be formed by a single kernel (unlike in traditional fuzzy sets of the so-called Mamdani, TS type, or relational fuzzy sets where the antecedent is an aggregation of fuzzy sets per input feature). The classification itself can be performed based on the well-known principle called ‘winner takes all’ which is often used in classification (Angelov and Zhou 2008b):

$$\text{Class} = \arg \max_{j=1}^C (\lambda_j). \quad (14)$$

It is important to note that this classifier is incremental. It is not evolving in the sense of (Angelov and Zhou 2008a, Angelov 2011) because the number of rules is fixed (equal to C), but is on-line. It will be evolving if new classes are added in a data stream. It is also important to note that this simplified FRB classifier is a typical incremental classifier that does not require an iterative training data set and a separate validation data set.

3.2 Parameters learning method

The total number of parameters for traditional FRB systems can be determined as $\text{TNP} = \text{NAP} + \text{NCP}$ (where TNP denotes total number of parameters, NAP is the number of antecedent parameters, and NCP is the number of consequent parameters). For traditional FRB with Gaussian scalar membership functions $\text{NAP} = 2 \times n \times N$ (where n is the number of input variables/features and N is the number of rules), TNP is equal to $N \times (n + 1)$. In total, a traditional FRB requires $N \times (3n + 1)$ parameters to be determined! According to the proposed concept, the antecedent part of the FRB system is *parameter free*. Therefore, $\text{NAP} = 0$. Although, the NCP is the same as for traditional FRB, the total number of parameters required is significantly (in *orders of magnitude!*) reduced which will be demonstrated on real industrial data in Section 5.

Therefore, parameter identification only involves learning the consequent part’s parameters. Once the antecedent part of the FRB system is determined, the identification of parameters of the consequent part, π^i , can be found as a recursive least square (RLS) estimation problem (Ljung 1999, Kailath *et al.* 2000). If we consider an on-line (or evolving) version, a number of additional issues must be addressed. These include:

- on-line normalization or standardization of the data streams and
- the real-time algorithm must perform both tasks (granulation and parameter estimation) at the same time instant (per data point) for a time significantly shorter than the sampling period.

Off-line standardization can be given by Hastie *et al.* (2001)

$$z = \frac{z^{\text{raw}} - \bar{z}}{\sigma},$$

where z^{raw} denotes raw not-standardized data vector.

The mean can be calculated recursively per scalar input/output $j = 1, 2, \dots, n + m$ by (7):

$$\bar{z}_{jk} = \frac{k-1}{k} \bar{z}_{j(k-1)} + \frac{1}{k} z_{jk}, \quad \bar{z}_{j1} = z_{j1}, \quad k = 2, 3, \dots \quad (15a)$$

The standard deviation can be calculated by Angelov and Zhou (2008a) and Angelov (2010),

$$\sigma_{jk}^2 = \frac{k-1}{k} \sigma_{j(k-1)}^2 + \frac{1}{k-1} (z_{jk} - \bar{z}_{jk})^2, \quad \sigma_{j1} = 0, \quad k = 2, 3, \dots \quad (15b)$$

While the antecedent part of the rules can be determined in a fully unsupervised way, the consequent part requires a supervised feedback. The supervision is in the form of error feedback, which guarantees optimality (subject to fixed rule base/NN structure) of the parameters of the consequent part.

The overall output of the simplified FRB system can be given in a vector form as follows:

$$y = \psi^T \theta \quad (16)$$

where $\theta = [(\pi^1)^T, (\pi^2)^T, \dots, (\pi^N)^T]^T$ is a vector formed by the sub-system parameters; $\psi = [\lambda^1 x_e^T, \lambda^2 x_e^T, \dots, \lambda^N x_e^T]^T$ is a vector of the inputs that are weighted by the normalized activation levels of the rules, λ^i , $i = [1, N]$ for the linear consequents, and $\psi = [\lambda^1, \lambda^2, \dots, \lambda^N]^T$ for the singleton type consequents.

For a given data point, x_k , the optimal in least square (LS) sense solution $\hat{\theta}_k$ that minimizes the following cost function:

$$(Y - \Psi^T \theta)^T (Y - \Psi^T \theta) \rightarrow \min \quad (17)$$

can be found applying weighted RLS, wRLS (Angelov 2010):

$$\hat{\theta}_k = \hat{\theta}_{k-1} + C_k \psi_k (y_k - \psi_k^T \hat{\theta}_{k-1}), \quad (18)$$

$$C_k = C_{k-1} - \frac{C_{k-1} \psi_k \psi_k^T C_{k-1}}{1 + \psi_k^T C_{k-1} \psi_k}, \quad (19)$$

where $\hat{\theta}_1 = 0$; C is a $Nn \times Nn$ co-variance matrix; $C_1 = \Omega I$, where Ω is a large positive number and I is the identity matrix; and $k = 2, 3, \dots$

wRLS is fuzzily weighted through the activation levels and is not the conventional weighted RLS which is directly applicable under the assumption that model (9) has a fixed structure. Under this assumption the optimization problem (19) is linear in parameters.

FRB classifiers can, generally, be of two types (Angelov and Zhou 2008b): (i) zero order when consequents of the rules constitute of the class labels and (ii) first order when the consequents of the rules are linear. For the former case, there are no parameters in the consequent part and for the latter case parameters can be found as described above

(Equations (16)–(19)). We will consider in Section 5 a numerical example of a one rule per class simplified FRB of zero order which is non-parametric.

4. Monitoring quality of Clouds

Monitoring the quality of the FRB structure and Clouds, in particular, is paramount for generating an effective structure. The quality of the Cloud can be characterized by their *age* and *utility*.

Each data sample is assigned to a Cloud at the moment it is first read by

$$M^i \leftarrow M^i + 1 \text{ for } i = \arg \max_{l=1}^N (\gamma^l), \quad i = [1, N]. \quad (20)$$

4.1 Age of the Cloud

An important quality measure that describes the properties of a Cloud is its *age*,

$$\text{age}_k^i = k - \bar{I}_j, \quad i = [1, N], \quad (21)$$

where I_j denotes the time index of the moment when the l th data sample was read; $\bar{I}_j = (1/(M_k^i)) \sum_{j=1}^{M_k^i} I_j$ is the mean time index of data samples which are associated with the i th Cloud.

The concept of Cloud *age* (see Figure 3 for an example) is specifically important for on-line models and systems and for real-time applications, which provides a compact measure of the dynamics of the data distribution and is spanned along the time domain.

Data density is a measure of the data distribution in the data space where the data points are timeless (stripped from their time tag). The *age* indicates how *old* is the information that supports certain Cloud and is thus of key importance for updating the FRB structure and detecting *concept drift* (Widmer and Kubat 1996, Angelov 2010) which corresponds to the inflexed point of the *age* curve (the point when the derivative of *age* in terms of time index, $(d(\text{age}))/dk$ changes its sign).

4.2 Utility

Utility (Angelov 2010) is associated with the whole fuzzy rule, not just the Granule (see Figure 4 for an example of the evolution of the utility of the two fuzzy rules that form the model).

It is defined as the accumulated firing level of a fuzzy rule given by Equation (3) summed over the life of each rule:

$$U_k^i = \bar{\lambda}^i, \quad i = [1, N], \quad (22)$$

where $\bar{\lambda}^i = (1/(k - I_i)) \sum_{j=I_i}^k \lambda_j^i$ denotes the mean utility.

Utility, U , accumulates the weight of the rule contributions to the overall output during the life of the rule (from the moment when this rule was generated till the current time instant, k). It is a measure of importance of the respective fuzzy rule comparing to the other rules. Utility can be used as a basis to simplify the rule base according to principle C, namely, to remove rules with low utility:

$$\text{IF}(U_k^i < \varepsilon_1) \text{ THEN } (\lambda^i \leftarrow 0), \quad i = [1, N], \quad (23)$$

where ε_1 is a small (up to 10%) tolerance threshold.

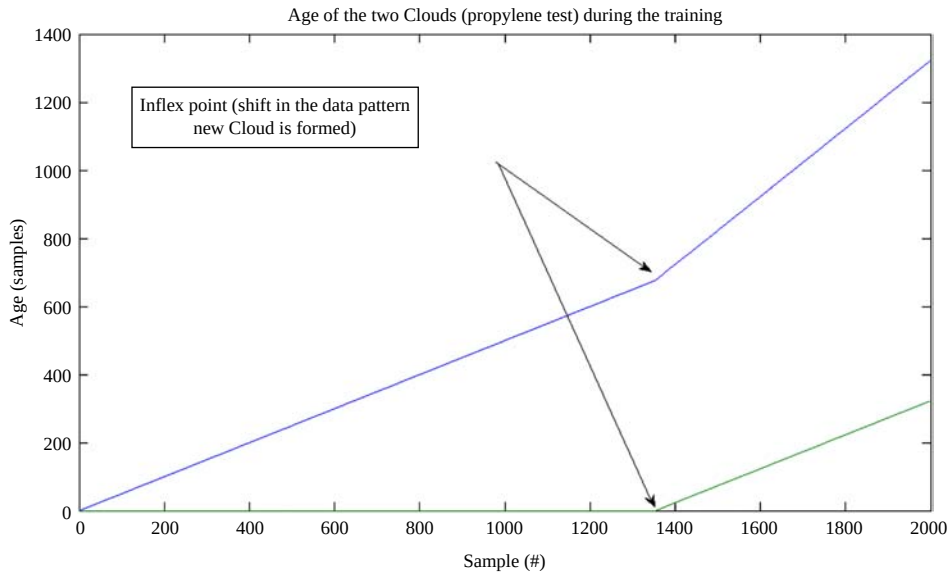


Figure 3. Evolution of the age of the Clouds (propylene experimental data).

4.3 Automatic selection of most relevant input variables

Selecting most informative input variables is an important pre-processing stage, which is often addressed by off-line approaches such as principle component analysis (PCA) (Hastie *et al.* 2001) and genetic programming (Kordon and Smits 2001). We propose to gradually remove input variables that contribute little to the output/s based on on-line estimation of the sensitivity of the output/s in terms of the inputs. Usually, we assume for the simplified FRB (Equation (9), the consequents to depend linearly on the inputs).

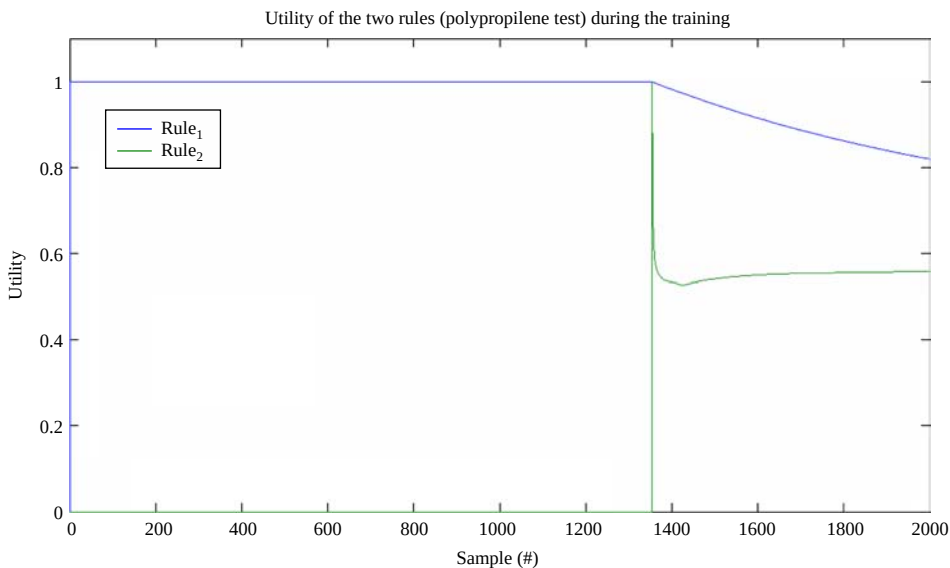


Figure 4. Evolution of the utility of the Clouds (propylene experimental data).

The importance of each input can be evaluated by the ratio of the accumulated sum of the consequent parameters for the specific j th input with respect to all n inputs (Angelov and Zhou 2008a, Angelov 2011):

$$\omega_{ijk} = \frac{T_{ijk}}{\sum_{r=1}^n T_{irk}}, \quad i = [1, N], \quad j = [1, n], \quad (24)$$

where $T_{ijk} = \sum_{l=1}^k |\alpha_{ijl}|$ denotes the accumulated sum of parameter values of the i th rule.

These weights can be used for a gradual removal of inputs/features, j^* that contribute little to the overall output (see Figure 8 for an example):

$$\exists j^* \mid \omega_{ijk} < \varepsilon_2 \max_{r=1}^n \omega_{irk}, \quad i = [1, N], \quad (25)$$

where ε_2 denotes the tolerable minimum weight of an input/feature – suggested value is 20%.

4.4 Procedure of the method

The dynamically evolving version of the proposed simplified data Clouds and density distribution based FRB approach can be very briefly presented by the following pseudo-code:

Begin

After initialization in real-time:
 Form **new Clouds** using (11) – (12);
 Monitor quality on-line and **remove Clouds** according to (22) – (23);
 Apply **wRLS**, (18) – (19) for existing and new Clouds
Select on-line the best inputs, (24) – (25)
 Repeat these steps for the next data sample ($k \leftarrow k + 1$) until no more data is available or until a requirement to stop the process.

End

4.4.1 Pseudo-code 1

It should be stressed again that the proposed approach is valid for all modes of operation such as off-line (possibly expert-based), on-line as well as evolving. It is also equally valid for prediction/estimation, classification as well as control applications of FRB. In this paper, only illustrative examples of the type of ‘proof of concept’ will be demonstrated while more detailed studies in each of the specific areas will be further considered in future publications (Figure 5).

5. Numerical examples

To test the newly proposed concept and method, we considered simple ‘proof of concept’ style examples with both predictive model and a classifier considering both evolving structure and fixed off-line case with incremental reading of the data samples. Recognizing the limitations of the demonstrative examples, we hope that future publications will cover more applications of this technique. For the evolving predictive model, we used one data stream from a well-known benchmark and two from real industrial processes. The overall performance of the proposed approach was analysed based on a comparison of the results

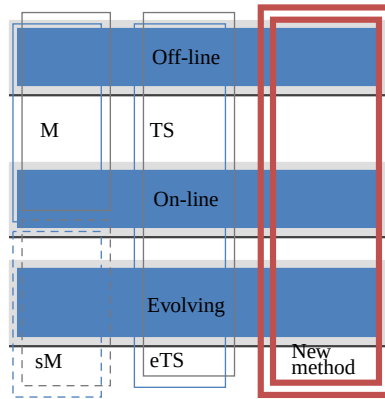


Figure 5. Different types of FRB systems: M, Mamdani; sM, simplified (singletons) Mamdani; TS, Takagi–Sugeno; new method, the proposed simplified FRB using data Clouds and density distribution. Note that each one of the off-line, on-line, and evolving versions also applies to prediction/estimation, classification, and control separately.

by applying more established techniques available in Matlab such as ANFIS (off-line; Jang 1993), *genfis2* (off-line; Yager and Filev 1993, Chiu 1994), DENFIS (evolving with off-line initialization; Kasabov and Song 2002), and eTS + (evolving; Angelov 2010). The main reason that these particular methods were used for comparison is that they are available as software widely and they were the cornerstone methods for automatic fuzzy and neuro-fuzzy systems design from data in an off-line and in an on-line, evolving manner.

In the test with ANFIS and *genfis2*, the data sets were separated into training and validation sub-sets. The training sets were used for the off-line training and the error in prediction was estimated based on the validation sub-sets.

5.1 Box–Jenkins gas furnace data

The Box–Jenkins data set is one of the well-established benchmark problems. It consists of 290 pairs of input–output data taken from a laboratory furnace (Box and Jenkins 1976). Each data sample consists of the methane flow rate, the process input variable, $u(k)$, and the CO₂ concentration in off gas, the process output, $y(k)$. From different studies, the best model structure for this system is

$$y(k) = f(y(k-1), u(k-4)). \quad (27)$$

The ‘trick’ is to determine a good (possibly non-linear) function, f (both in terms of its structure and parameters). Obviously, the number of input variables is 2. Traditionally, off-line models use 200 data samples for training and 90 for validation. Evolving models (such as DENFIS, eTS +, or the proposed new method) do not need to separate training and validation data in principle, but we did this in this experiment primarily to put these models on the same footing with the off-line counterparts. The values of the performance measures were calculated for the validation data. The so-called non-dimensional error index (NDEI) defined as the ratio of the root mean square error (RMSE) over the standard deviation of the target data was used to compare model performance as well as the RMSE itself. The results are tabulated in Table 3 (the time is shown per sample).

Table 3. Box–Jenkins gas furnace data.

Method	ANFIS	Genfis2	DENFIS	eTS+	New
Type	Off-line		Evolving		
RMSE	0.100	0.050	0.052	0.047	0.043
NDEI	0.605	0.311	0.322	0.291	0.272
# rules	25	3	10	7	7
#parameters	175	21	70	49	21
# inputs	2	2	2	2	2
Time (ms)	–	–	3.1	3.4	2.7

Note: Values in bold indicate best values.

From Table 3 it is seen that using the proposed new method a *simple and compact* fuzzy model of seven fuzzy rules can be extracted from this data stream with significantly smaller number of parameters and better precision. For example, rule 1 derived by this method is

Rule¹ : IF (x is like Cloud¹) THEN $\left(y_k^1 = 0.4008 + \begin{bmatrix} -0.4135 \\ 0.6061 \end{bmatrix} \begin{bmatrix} y_{k-1} & u_{k-4} \end{bmatrix} \right)$.

Note that there is no need to define Gaussian or triangular membership functions for the antecedent part and the likeness of a particular input data sample is judged by its local density (4a) and (6), i.e. by the closeness to *all* data samples of that Cloud, not only to its centre (which is also not required to be defined). The antecedent part also does not require parameters (such as spread or apex points to be defined and updated).

5.2 Propylene case study

The propylene data set is collected from a chemical distillation process run at The Dow Chemical Co., TX (USA) [courtesy of Dr A. Kordon (Angelov and Kordon 2010)]. The data set consists of 3000 readings from 23 sensors that are on the plant. They are used to predict the propylene content in the product output from the distillation. Some of the inputs may be irrelevant to the model and thus bring noise. Therefore, the input selection is very important task, which is usually done off-line as a part of the pre-processing. Instead, the procedure proposed in this paper leads to an effective selection of most relevant inputs (in this case, the best input variable is x_8). The results (tabulated in Table 4) illustrate that

Table 4. Polypropylene data.

Method	ANFIS	Genfis2	DENFIS	eTS+	New
Type	Off-line		Evolving		
RMSE	Cannot cope with dimensionality (crash, memory full)			0.157	0.137
NDEI				0.444	0.388
# rules	N	N	N	6	2
#parameters	$70 \times N$	$70 \times N$	$70 \times N$	38	8
# inputs	23	23	23	2	1
Time (ms)	–	–	–	2.38	1.44

Note: Values in bold indicate best values.

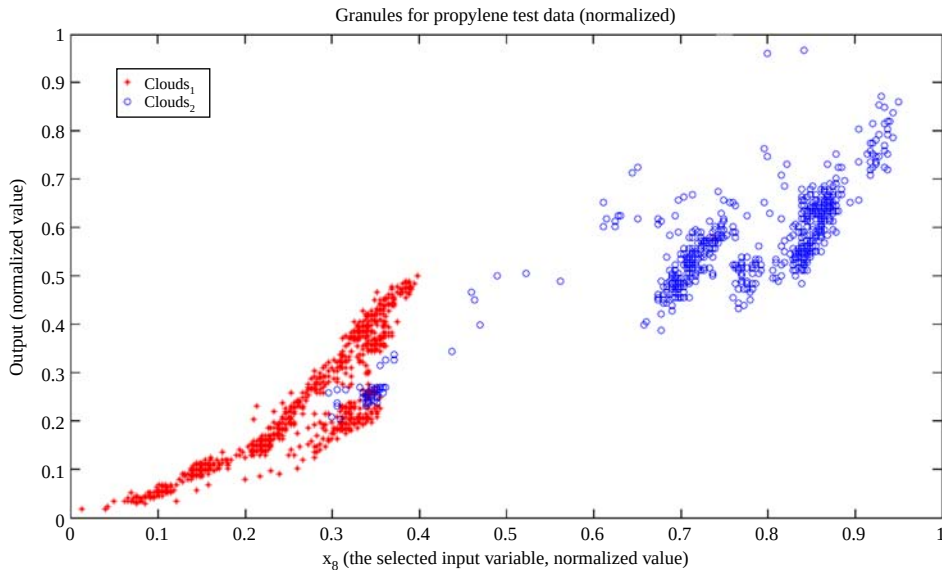


Figure 6. Clouds for the propylene data. The figure illustrates that using traditional Gaussian or triangular membership functions is far from the real data distributions. It is easy to note that for this data distribution an Euclidean type circular shape cluster or even an ellipsoidal (Mahalanobis) type advanced clustering will fail to correctly and fully represent the real data distribution.

highly compact FRB system which consists of only two fuzzy rules can be extracted from a data stream automatically and this simplified FRB system (intelligent sensor) can model the propylene from a real (noisy) data stream with very good precision.

The two fuzzy rules have the following linguistic description that demonstrates the high transparency of the proposed type of FRB systems:

Final Rule-base for Propylene:

Rule ¹: **IF** (x_8 is like Cloud ¹) **THEN** ($y^1 = -0.01 + 0.80x_8$)

Rule ²: **IF** (x_8 is like Cloud ²) **THEN** ($y^2 = -0.14 + 0.942x_8$)

In this case, also there is no need to define Gaussian or triangular membership functions for the antecedent part but one can extract scalar membership function for both fuzzy rules (see Figure 4). Note that this is not necessary for the computations on which the approach is based, but is for illustrative purposes. The two Clouds that were formed automatically are depicted in Figure 6.

It is obvious from Figure 6 that a Gaussian or even triangular or trapezoidal membership functions would have been a gross simplification of the real data distribution which is taken fully into account in the proposed approach. In Figure 3, the 'ageing' of the Clouds is demonstrated and in Figure 4, the evolution of the Utility of both fuzzy rules is demonstrated.

5.3 NOx emissions modelling

This data (courtesy of Dr E. Lughofer, University of Linz, Linz, Austria) describe testing car engines for modelling the NOx emissions in their exhausts. In this experiment, initially

Table 5. NOx emission data.

Method	ANFIS	Genfis2	DENFIS	eTS+	New
Type	Off-line			Evolving	
RMSE				0.057	0.054
NDEI	Can not cope with huge dimensionality (crash, memory full)			0.324	0.309
# rules	N	N	N	9	1
#parameters	$541 \times N$	$541 \times N$	$541 \times N$	1170	44
# inputs	180	180	180	43	43
Time(ms)	–	–	–	11.5	2.95

180 different physical variables were measured using conventional (hard) sensors. Most of these potential inputs to the regression model are the same physical variables (such as pressure in the cylinders, engine torque, and rotation speed) measured at different time instances (different time delays). The aim is to evaluate the content of NOx in the emissions from the exhaust automatically. The proposed method is used to generate automatically and evolve a nonlinear regression model form the input data streams. The performance was compared to the alternative methods but due to huge dimensionality only eTS+ was able to cope (see Table 5 for the results).

The problem of selecting a small subset of the highly correlated inputs is not easy and is usually done subjectively based on the *prior* knowledge, experience, or off-line methods. Both eTS+ and the new simplified FRB were able to automatically select a smaller subset of inputs (43) as demonstrated in Figure 7. However, the newly proposed method requires over 20 times smaller number of parameters yet providing a better performance (precision, time, and number of rules).

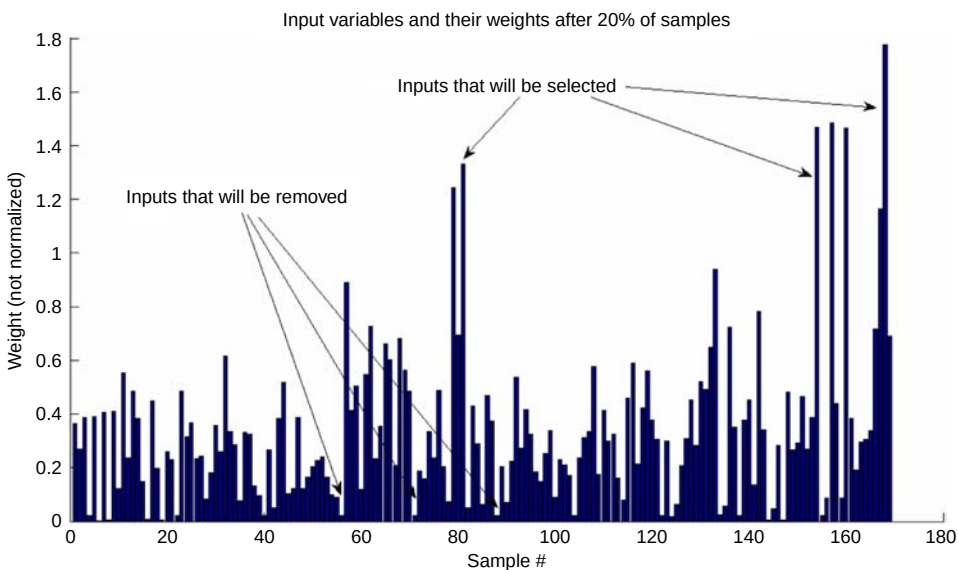


Figure 7. Illustration of the input variables selection process for the NOx data set. Top plot – features down selected from the initial 180 after 20% of the samples (after 165 samples). Bottom plot – features down selected finally (after all samples have been used).

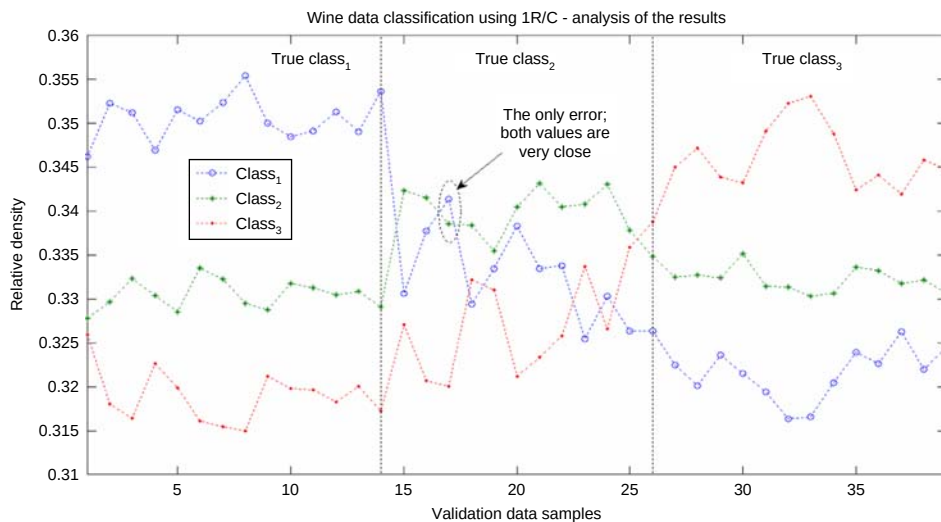


Figure 8. The proposed simple FRB classifier takes the maximum of the relative density (14) and determines the winning class label. Note the closeness of the values in the validation case 17 (the only one that was misclassified).

5.4 Simple non-parametric FRB classifier

In this paper, we used a ‘proof of concept’, the so-called wine data set (University of California at Irvine (UCI) 2010). It contains data from a chemical analysis of wines grown in the same region of Italy but derived from three different cultivars (thus, three class labels). The analysis determined the quantities of 13 constituents found in each of the three types of wines (the 13 input features/attributes of the classifiers). Figure 8 depicts the results as well as the mechanism based on which the classifier works.

It is interesting to note not only a very high classification rate (97.44% correct classification), but also the marginal difference between the relative density for the validation sample 17 which was the only sample to be misclassified. The values of the relative density, λ , can be provided to the decision maker and can indicate possible problematic cases for further investigation or declaring ‘not sure’ outcome. It is important to note that the proposed classifier is computationally very light and recursive. Table 6 provides a numerical comparison of the results achieved by the proposed simple FRB classifier and other classifiers for the same data set.

Table 6. Comparative results of various classifiers for wine data set.

Classifier	Classification rate (%)	# of rules
Proposed	97.44	3
kNN	96.94	3 ^a
eClass 0 (Angelov and Zhou 2008a)	92.44	9.94 ^b
eClass 1 (Angelov and Zhou 2008a)	97.22	6.4 ^c
C4.5	92.13	4.6

^a kNN does not provide any insight of how the result is achieved (model structure) and does not take into account all data as the newly proposed classifier does.

^b The rules of eClass 0 have much more complex consequent.

^c The rules of eClass 1 have much more complex antecedent and consequent.

Values in bold indicate best values – maximum for classification rate and minimum for the number of rules.

6. Conclusion and future direction

In this paper, we proposed an alternative type of FRB which goes further in the conceptual and computational simplification of the antecedent part while preserving the best features (flexibility, richness combined with simplicity, and modularity) of its predecessors (Mamdani and TS type FRB). The newly proposed concept of complex systems and functions is seen as the next more efficient form of system modelling applicable to time-series prediction, clustering, classification, control, decision support systems, and other problems where conventional fuzzy rule-based systems are used. It has a non-parametric form that reflects *all* data (instead of attempting to approximate them with parametric functions, e.g. Gaussian, triangular, trapezoidal, etc., as the conventional systems do). The main advantages of the new method are that while keeping all the advantages of ‘traditional’ FRB systems they avoid the well-known problems related to (scalar) membership functions definition, identification, and update. It takes into account the spatial distribution and similarity of *all* the data by proposing an innovative and much simplified form of the antecedent part. The proposed concept is applicable to off-line, on-line, and evolving (dynamic structure) types of FRB system design. An example of an evolving simplified FRB system is presented for prediction but equally the approach is applicable to fuzzy classifiers and controllers design – something that will be further studied and published.

Note

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Notes on contributors



Dr Plamen Angelov is a Reader in Computational Intelligence and coordinator of the Intelligent Systems Research Area (which includes 8 academics as well as over 30 Research Associates (RAs) and PhD students with a portfolio of over £1M), within the School of Computing and Communications which is based in Infolab21. He received MEng (1989) and PhD (1993) degree from Sofia Technical University and Bulgarian Academy of Sciences (BAS) respectively and spent ten years as a research fellow in BAS, University of Leuven-la-neuve, Belgium, Loughborough University, UK prior to joining Lancaster University in 2003 as a Lecturer. He held Visiting Professor positions in various Universities (Campinas, Brazil - 2005; University of Wolfenbutel, Germany 2007; Carlos III, Madrid, Spain - 2010).

He is Chairing two Technical Committees (TC) of IEEE - on Standards with Computational Intelligence Society and on Evolving Intelligent Systems with Systems, Man and Cybernetics Society. He is a co-recipient of several best paper awards at IEEE conferences (2006 and 2009) and of two prestigious Engineer 2008 Technology + Innovation awards for Aerospace and Defence and the Special Award. Dr Angelov is Editor-in-Chief of the Springer journal *Evolving Systems* (ISSN 1868-6478) and Associate Editor (AE) of prestigious *IEEE Transactions on Fuzzy Systems* and of Elsevier's *Fuzzy Sets and Systems* journal as well as AE of several other journals in the area of computational intelligence. He was the General Chair of a number of IEEE conferences during last five years, including the annual IEEE Symposium on Evolving and Adaptive Intelligent Systems and the premier event in the area of neural networks – International Joint Conference on Neural Networks (IJCNN) in 2013 which will be held in Dallas, TX, USA. Dr Angelov is regularly invited to join International Programme Committees of prestigious IEEE, IFAC, IFSA etc. conferences as well as to give key note, plenary and invited talks at prestigious conferences, leading companies and

events. He also regularly organises tutorials, special sessions and sits on panels at leading IEEE conferences.

Dr Angelov is a prolific author with high impact publications. He authored or co-authored well above 150 publications including over 50 peer reviewed journal papers including many IEEE Transactions articles, high impact papers in journals such as Nature protocols, Analyst, etc. He also authored six books, including one monograph (Springer, 2002) and second being accepted (to appear in 2012 by Wiley). He authored three patent applications one of which was licensed to the Global giant Ford Motor Co. (2011) and used in a refinery (CEPSA) in Spain, in the chemical plants of The Dow Chemical, TX, USA and other companies. His papers are highly cited (overall they collect well over 1800 citations with his most cited paper alone collecting over 280 citations making it one of the 0.01% most cited publications in Computing and Engineering areas according to ISI World of Science; his so called 'h-index' is 19 with over 100 citations pa and over 10 publications pa on average).

Dr Angelov holds a portfolio of research projects and attracted since he joined Lancaster University well over £1M of research funding (over £160K pa for the last five years; over £120K pa if take into account the Principle Investigator, PI/co-investigator(s), co-I(s) split). He was awarded in total over a dozen research projects, some of which were very large consortia (e.g. £32M ASTREA, £9M GAMMA, £1.3M SVETLANA) were the above mentioned figures are the share of Lancaster University. For the last five years he was awarded on average about 2 projects pa with source of funding including EPSRC, EU FP7, MoD, DTI/BIS, industry (BAE Systems), The Royal Society, etc.

Dr Angelov has currently eight PhD students (four of which are in writing up stage) and two RAs and four awarded PhDs. In addition he regularly hosts visiting PhD students (from Spain, Slovenia), postdocs (Slovenia, Austria) and professors (Germany) funded by The Royal Society or their home research agencies. In the past Dr Angelov supervised half a dozen other RAs. He supervised several dozens of Master and undergraduate students many of whom received distinction and prestigious awards (IEEE, Nokia) and published their first publications at prestigious IEEE events before or just after their graduation. He is regularly invited to serve as external examiner in Universities around the world, including Oxford, Barcelona, Patras, Auckland, Seville, Essex, Leicester, London. Dr Angelov was invited to review research project proposals by various research organisations from UK, Canada, Austria, Greece, Bulgaria.

The research activity of Dr Angelov has been publicised in the prestigious IEEE Magazine (2009), Aviation Week (2009), Flight Global (2008), Airframer (2007), Lancaster University Annual Report (2011, p.43) and other journals (Fuzzy Sets and Systems, 1999) and outlets (EUNITE, 2001).



Ronald R. Yager has worked in the area of machine intelligence for over twenty-five years. He has published over 500 papers and fifteen books in areas related to fuzzy sets, decision making under uncertainty and the fusion of information. He is among the world's top 1% most highly cited researchers with over 7000 citations. He was the recipient of the IEEE Computational Intelligence Society Pioneer award in Fuzzy Systems. Dr. Yager is a fellow of the IEEE, the New York Academy of Sciences and the Fuzzy Systems Association. He was given a lifetime achievement award by the Polish Academy of Sciences for his contributions. He served at the National Science Foundation as program director in the Information Sciences program. He was a NASA/Stanford visiting fellow and a research associate at the University of California, Berkeley. He has been a lecturer at NATO Advanced Study Institutes. He is a visiting distinguished scientist at King Saud University, Riyadh Saudi Arabia. He is a distinguished honorary professor at the Aalborg University Denmark. He is an affiliated distinguished researcher at

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